



Figure 6.2. Example of a graph requiring extra combinatoric factors in the S -matrix. For an interaction involving, say, three factors of some field (as well as other fields) we usually include a factor $1/3!$ in the interaction Hamiltonian, to cancel factors arising from sums over ways of pairing these fields with their adjoints in other interactions. But in this diagram there are two such factors of $1/3!$ and only $3!$ different pairings, so we are left with an extra factor of $1/3!$.

can be paired with any one of the M identical fields in $\mathcal{H}_i(x)$; the second with any one of the remaining $M - 1$ identical fields; and so on, yielding an extra factor of $M!$. To compensate for this, it is conventional to define the coupling constants g_i so that an explicit factor $1/M!$ appears in any $\mathcal{H}_i(x)$ containing M identical fields (or field adjoints.) For instance, the interaction of M th order in a scalar field $\phi(x)$ would be written $g\phi^M/M!$. (More generally, one often also displays an explicit factor of $1/M!$ when the interaction involves a sum of M factors of fields from the same symmetry multiplet, or when for this or any other reason the coupling coefficient is totally symmetric or antisymmetric under permutations of M boson or fermion fields.)

However, this cancellation of $M!$ factors is not always complete. For instance, consider a Feynman diagram in which the M identical fields in one interaction $\mathcal{H}_i(x)$ are paired with M corresponding field adjoints in a *single* other interaction $\mathcal{H}_j(y)$. (See Figure 6.2.) Then by following the above analysis, we find only $M!$ different pairings (since it makes no difference which of the field adjoints we call the first, second, ...), cancelling only one of the two factors of $1/M!$ in the two different interactions. In this case, we would have to insert an extra factor of $1/M!$ 'by hand' into the contribution of such a Feynman diagram.

Other combinatoric factors arise when some of the permutations of vertices have no effect on the Feynman diagram. We noted earlier that the factor $1/N!$ in the series (6.1.1) is usually cancelled by the sum over the $N!$ diagrams that differ only in the labelling of the N vertices. However, this cancellation is incomplete when relabelling the vertices does not yield a new diagram. This happens most commonly in the calculation of vacuum-to-vacuum S -matrix elements in a theory with a quadratic interaction $\mathcal{H} = \psi^\dagger M \psi$, where M may depend on external fields. (The physical significance of such vacuum fluctuation diagrams is discussed in detail in Volume II.) The Feynman diagram of N th order in $\mathcal{H}$ is a